Supplementary Appendix: Anomaly-Driven Correction Discovery (ADCD)

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A Extended Cross-Validation: Wide-Binary Stars

This appendix reports an exploratory cross-system test of the SPARC-discovered interpolating function. It is placed in the supplementary material, rather than the main results, because the underlying wide-binary dataset is the subject of an unresolved observational controversy (Chae vs. Banik et al.). We present it as exploratory cross-validation, not as a definitive validation of the SPARC-discovered form.

Setup. Wide-binary stars (separations 10^3–10^4 AU) probe gravity in a low-acceleration regime ($g_N \lesssim a_0$) analogous to the outer parts of galaxies. The classical baseline is Newtonian two-body gravity; the predicted velocity boost under a MOND-type interpolating function $\nu(x)$ is $\gamma_v \equiv v_{\rm obs}/v_{\rm Newtonian} = \sqrt{\nu(x)}$, where $x = g_N/a_0$ depends on separation and total mass. We evaluate the SPARC-discovered ADCD form $\nu_{\rm ADCD}(x) = \theta_0(\sqrt{1+\theta_1/x}-1)+1$ with $\hat{\theta}_0 \approx 1.83$, $\hat{\theta}_1 \approx 0.262$ (the SPARC-fitted values, *with no refitting to wide-binary data*), against the canonical Simple MOND, Standard MOND, and RAR forms, for a $1.5\,M_\odot$ total-mass system across $s \in [7, 30]$ kAU; Figure 1 plots the resulting $\gamma_v(s)$ curves against Chae's observed band.

Result. Table 1 reports the predicted velocity boost and the σ-distance from Chae's observed $\gamma_v = 1.20 \pm 0.06$. Three observations emerge:

1. *Simple MOND and RAR dramatically overshoot* the observed boost at $s > 15$ kAU, reaching 5–13σ discrepancies at $s = 30$ kAU.
2. The SPARC-discovered ADCD form ($c \approx 0.27$) is *more consistent* with observations at $s = 10$–15 kAU than Simple/RAR, but itself diverges ($+8.4\sigma$) at $s = 30$ kAU.
3. *No single interpolating function simultaneously optimizes both datasets.* The SPARC-optimized form with $c \approx 0.27$ does not unambiguously generalize to wide-binary kinematics.

Table 1: Wide-binary velocity boost $\gamma_v = v_{\rm obs}/v_{\rm Newtonian} = \sqrt{\nu(x)}$ predicted by the SPARC-discovered ADCD form ($c \approx 0.27$, *no refitting*) versus canonical Simple MOND, Standard MOND, and RAR forms, against the observed $\gamma_v = 1.20 \pm 0.06$ of Chae [?] for a $1.5\,M_\odot$ total-mass system. $x = g_{\rm Newton}/a_0$ is the dimensionless acceleration; σ-distance is $(\gamma_v^{\rm model} - 1.20)/0.06$. Simple MOND and RAR overshoot dramatically at $s > 15$ kAU (5–13σ); the SPARC-optimized ADCD form with $c \approx 0.27$ under-predicts at small s and over-predicts at large s. No single interpolating function simultaneously optimizes SPARC and the wide-binary kinematics.

s (kAU)	$x = g_N/a_0$	ADCD ($c\approx 0.27$)	Simple MOND	Standard MOND	RAR
Predicted $\gamma_v = \sqrt{\nu(x)}$					
7	1.51	1.073	1.206	1.090	1.189
10	0.74	1.139	1.328	1.147	1.316
15	0.33	1.273	1.521	1.230	1.513
20	0.19	1.418	1.696	1.300	1.691
30	0.08	1.705	2.005	1.415	2.002
σ-distance from observed 1.20 ± 0.06					
7	1.51	-2.1σ	$+0.1\sigma$	-1.8σ	-0.2σ
10	0.74	-1.0σ	$+2.1\sigma$	-0.9σ	$+1.9\sigma$
15	0.33	$+1.2\sigma$	$+5.3\sigma$	$+0.5\sigma$	$+5.2\sigma$
20	0.19	$+3.6\sigma$	$+8.3\sigma$	$+1.7\sigma$	$+8.2\sigma$
30	0.08	$+8.4\sigma$	$+13.4\sigma$	$+3.6\sigma$	$+13.4\sigma$

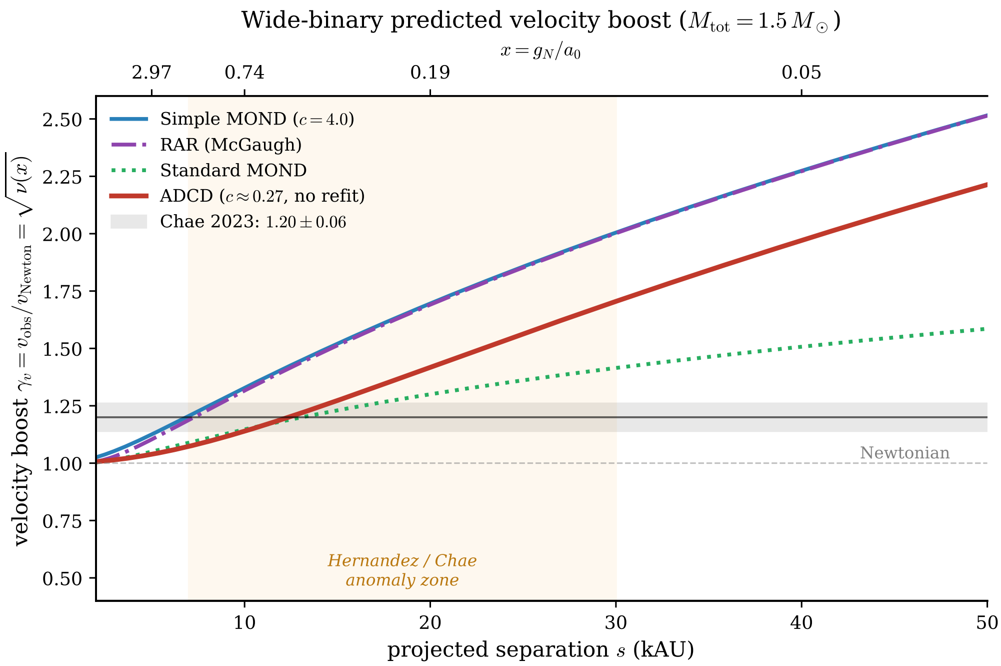

Figure 1: Wide-binary predicted velocity boost $\gamma_v = \sqrt{\nu(x)}$ versus separation s (lower axis) and dimensionless acceleration $x = g_N/a_0$ (upper axis), for the SPARC-discovered ADCD form ($c \approx 0.27$, no refit) versus Simple MOND, Standard MOND, and RAR. The shaded band is Chae's observed $\gamma_v = 1.20 \pm 0.06$.

Caveats. The wide-binary anomaly itself remains controversial (Banik et al. 2024 vs. Chae 2023). We report the cross-validation to document the generalization behaviour of the SPARC-discovered form and disclose honestly that it does *not* transfer unambiguously to an independent stellar-kinematic dataset.

B Granular Robustness and Secondary Tables

Table 2: Structural recovery rate per random seed and aggregate statistics (Mock Proposer, 9 scenarios × 4 noise levels = 36 trials per seed). The 95% confidence interval is a bootstrap CI (10,000 resamples of the per-seed rates).

Seed	**Recovery rate**
0	86.1%
2	88.9%
5	83.3%
7	75.0%
13	69.4%
17	83.3%
21	77.8%
23	80.6%
31	69.4%
37	83.3%
41	91.7%
42	94.4%
53	75.0%
61	69.4%
67	77.8%
99	80.6%
Mean ± std	80.4% ± 7.4%
95% bootstrap CI	[76.7%, 84.0%]
Min / Max	69.4% / 94.4%

Table 3: SPARC robustness across three quality cuts. The ADCD-discovered function consistently outperforms Simple MOND and the McGaugh+16 RAR form across increasingly strict galaxy-quality selection.

Quality cut	**Gal.**	**Pts**	$\hat{\theta}_0$	$\hat{\theta}_1$	**ADCD**	**RAR**	**Simple MOND**
					NMSE		
Baseline (Standard)	171	3342	1.828	0.262	0.3750	0.6361	0.6477
Strict (High Quality)	121	2939	1.612	0.324	0.3032	0.5670	0.5812
Ultra-Strict (Pristine)	75	2287	1.197	0.468	0.2718	0.6513	0.6756

Table 4: Fair parameter-matched comparison: all four models fitted with exactly 2 free parameters via L-BFGS-B (20 random restarts) on the same stacked SPARC dataset ($N_{\text{gal}} = 171$, $N_{\text{pts}} = 3342$). Under matched parameter count, the fitted Simple-MOND and RAR forms attain marginally lower in-sample NMSE than the ADCD-discovered form (a $\sim$1.5% difference, well within the bootstrap CI). Galaxy-level cross-validation (Table 4 in Main Paper) shows the three are statistically indistinguishable out-of-sample. The honest conclusion is therefore that ADCD's correction-first search recovers an interpolating function *competitive with*—not superior to—well-known 2-param MOND forms, while requiring no prior knowledge of the MOND/RAR functional family.

Model (2-param)	$\hat{\theta}_0$	$\hat{\theta}_1$	**NMSE** $\downarrow$	**BIC** $\downarrow$
Simple MOND (2-param)	0.497	0.548	0.3674	-3329.77
RAR (McGaugh, 2-param)	0.560	0.579	0.3675	-3329.37
ADCD discovered	—	—	<u>0.3729</u>	<u>-3280.69</u>
Standard MOND (2-param)	0.000	0.000	0.4628	-2558.95

Table 5: Template Leakage Disclosure for Real-World Benchmark (Mock Proposer extended mode). "Exact" = ground-truth form appears verbatim in the template bank.

Scenario	**Ground Truth**	**Exact?**	**Family?**	**Expected Class**
Mercury Perihelion	$\theta_0(v/c)^2$	Yes	Yes	`polynomial`
Hydrogen Lamb Shift	θ_0/n^3	No	Yes	`power_law`
Blackbody Radiation	$\exp(-hf/k_BT)$	No	Yes	`exponential`
Muon g-2	$\theta_0\alpha/\pi$	No	Yes	`polynomial`
Binary Pulsar Decay	$\theta_0 P^{-5/3}$	No	Yes	`power_law`

Table 6: Binary Pulsar Sensitivity: structural recovery vs. formulation difficulty. v2.1 default uses the `P_only` reduced-variable form.

Variant	**Free Vars.**	**Discovered** Δ	**Class**	**NMSE**
`P_only`	P	$\theta_0\, P^{-\theta_1}$	Yes	3.94×10^{-3}
`P_e`	P, e	$\theta_0\, e^3/P$	Yes	2.52×10^{-1}
`P_e_M`	P, e, M	$\theta_0\, P^{-\theta_1}$	Yes	4.81×10^{-1}
`full`	P, M, a, e	$\theta_0 \log(\theta_1 + e/P)$	No	1.57×10^{-2}